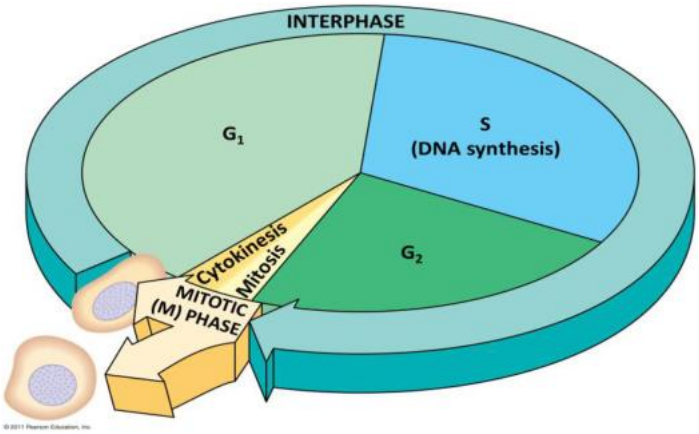
Medicine, Dentistry and Medical laboratory

Cell Biology

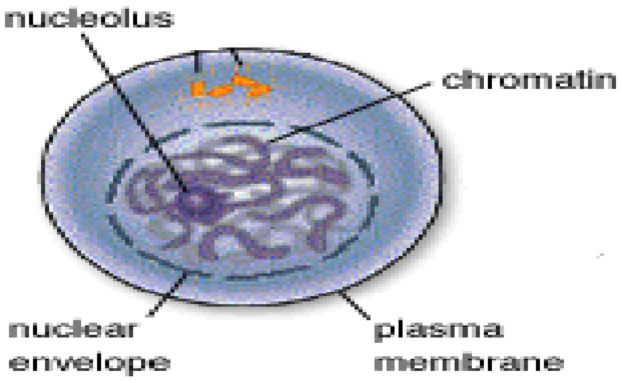
Lecture 8&9

The Cell Cycle



The **cell cycle** is an orderly set of stages that take place between the time a eukaryotic cell divides and the time the resulting daughter cells also divide.

(A) Interphase:



* Most of the cell cycle is spent in **interphase.** This is the time when a cell performs its usual functions, depending on its location in the body.

* In the past, **interphase** was known as the **resting stage**.

* **Interphase** consists of three stages: **G1, S,** and **G2.**

1- G1 Stage:

* During **G1**, the cell recovers from the previous division. Then, the cell increases in size, doubles its organelles and accumulates materials that will be used for **DNA** synthesis.

* Otherwise, during **G1**, cells are constantly performing their normal daily functions, including communicating with other cells, secreting substances, and carrying out cellular respiration.

2- S Stage:

* Following **G1**, the cell enters the **S stage**, when **DNA** synthesis or replication occurs. At the beginning of the **S stage**, each chromosome is composed of one **DNA** double helix.

* Following **DNA** replication, each chromosome is composed of two identical **DNA** double helix molecules. Each double helix is called a **chromatid.**

3- G2 Stage:

* Following the **S stage**, **G2** is the stage from the completion of **DNA** replication to the onset of mitosis. During this stage, the cell synthesizes proteins that will assist cell division.

* For example, it makes the proteins that form microtubules.

* Microtubules are used during the mitotic stage to form the mitotic spindle.

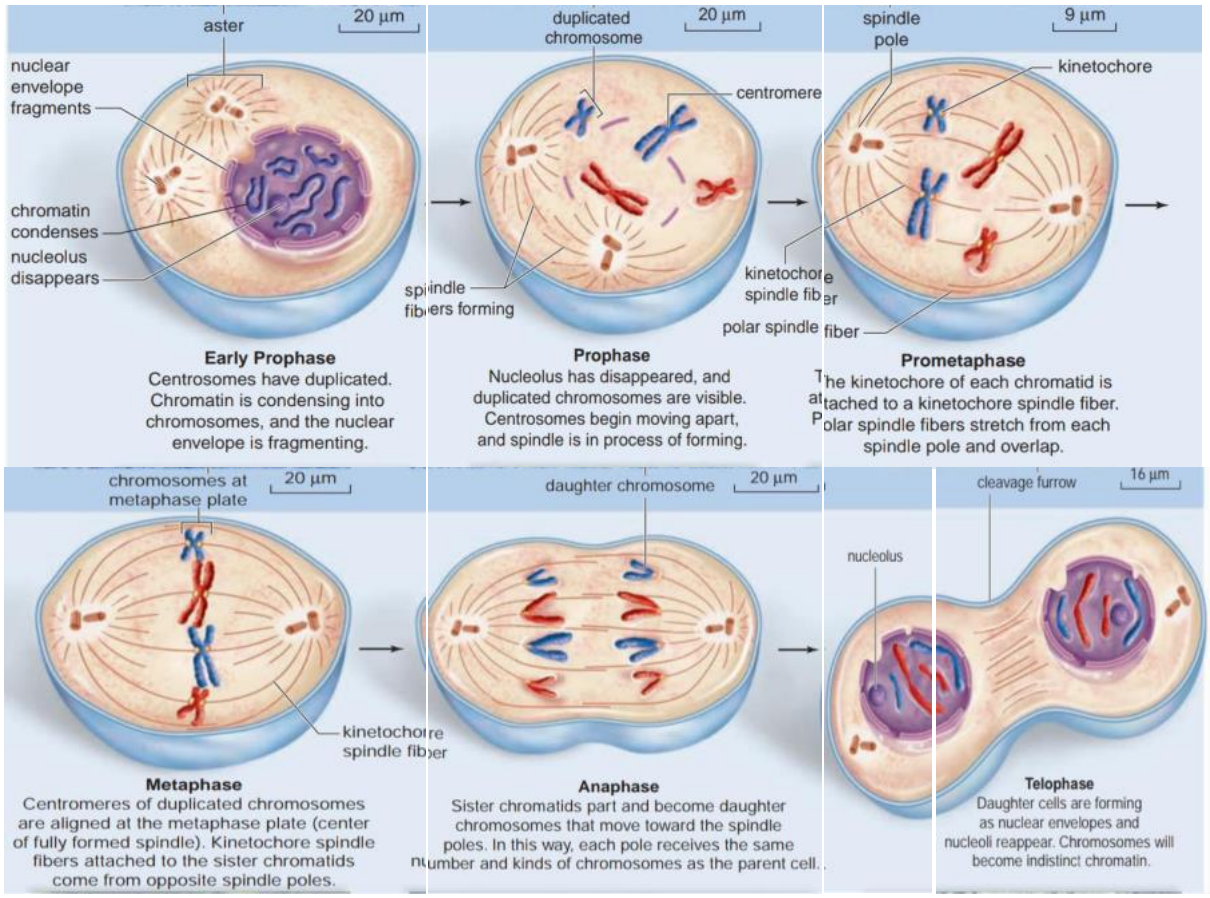
(B) Mitotic (M) Stage:

* Following **interphase**, the cell enters the **M stage**. This cell division stage includes **mitosis** (nuclear division) and **cytokinesis** (division of the cytoplasm).

Phases of Mitosis:

(I) Prophase.
(II) Metaphase.
(III) Anaphase.
(IV) Telophase.

Phases of Mitosis:



(I) prophase:

* During prophase, chromatin has condensed and the chromosomes are visible, the nucleolus disappears and the nuclear envelope fragments. The spindle begins to assemble as the two centrosomes migrate away from one another.

* In animal cells, an array of microtubules radiates toward the plasma membrane from the centrosomes. These structures are called **asters.** It is thought that asters serve to brace the centrioles during later stages of cell division. Notice that the chromosomes have no particular orientation because the spindle has not yet formed.

(II) Metaphase:

* During **metaphase,** the centromeres of chromosomes are now in alignment on a single plane at the center of the cell.
* The chromosomes usually appear as a straight line across the middle of the cell when viewed under a light microscope.

* The kinetochores of each chromosome are attached properly to spindle fibers and the chromosomes are properly aligned along the metaphase plate.

(III) Anaphase:

* At the start of **anaphase,** the two sister chromatids of each duplicated chromosome separate at the centromere, giving rise to two daughter chromosomes.

* Daughter chromosomes, each with a centromere and single chromatid composed of a single double helix, appear to move toward opposite poles.

* Anaphase is the shortest phase of mitosis.

(IV) Telophase:

* During **telophase,** the spindle disappears as new nuclear envelopes form around the daughter chromosomes.

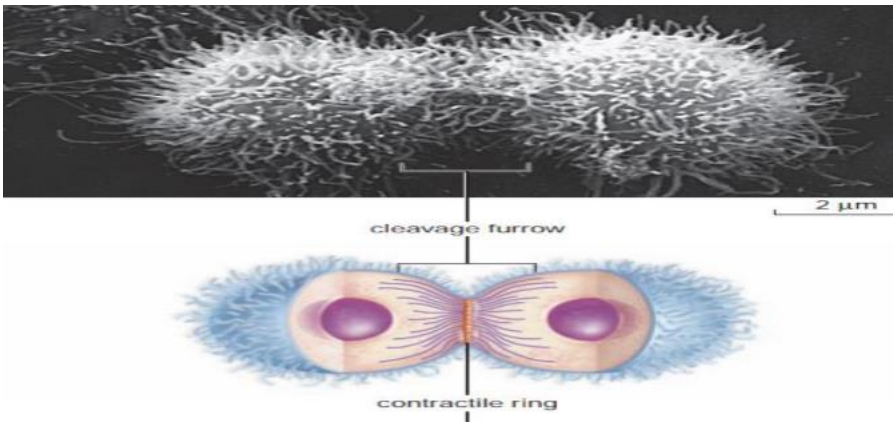
* Each daughter nucleus contains the same number and kinds of chromosomes as the original parent cell.

* Remnants of the polar spindle fibers are still visible between the two nuclei.

* The chromosomes become more diffuse chromatin once again, and a nucleolus appears in each daughter nucleus.

3- Cytokinesis:

* **In animal cells**, a **cleavage furrow**, which is an indentation of the membrane between the two daughter nuclei, forms just as anaphase draws to a close.



* By that time, the newly forming cells have received a share of the cytoplasmic organelles that duplicated during the **previous** interphase.

* The **cleavage furrow** deepens when a band of actin filaments, called the **contractile ring**, slowly forms a circular constriction between the two daughter cells and continues to separate the cytoplasm until there are two independent daughter cells.
* Cytokinesis **in plant cells** involves the building of new cell walls between the daughter cells.
* The **cell plate** is simply newly formed plasma membrane that expands outward until it reaches the old plasma membrane and fuses with this membrane. The new membrane releases molecules that form the new plant cell walls.
* These cell walls, known as primary cell walls, are later strengthened by the addition of cellulose fibrils. The space between the daughter cells becomes filled with middle lamella, which cements the primary cell walls together.

The Functions of Mitosis

* Mitosis permits growth and repair.
* In both plants and animals, mitosis is required during development as a single cell develops into an individual.
* In plants, the individual could be a fern or daisy, while in animals; the individual could be a grasshopper or a human being.
* In human beings and other mammals, mitosis is necessary as a fertilized egg becomes an embryo and as the embryo becomes a fetus.
* Mitosis also occurs after birth as a child becomes an adult.
* Throughout life, mitosis allows a cut to heal or a broken bone to mend.

Meiosis

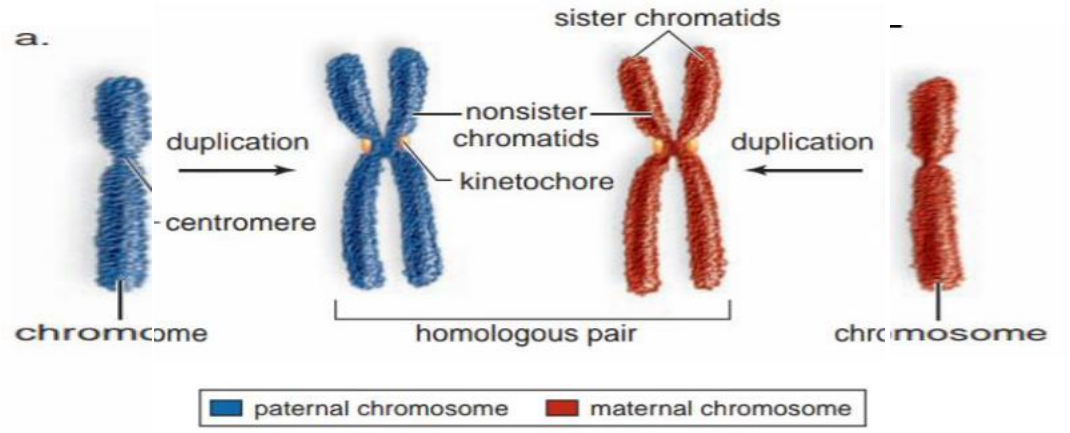
(I) Halving the Chromosome Number:

* In sexually reproducing organisms, **meiosis** is the type of nuclear division that reduces the chromosome number from the diploid (2n) number to the haploid (n) number.
* The **diploid (2n) number** refers to the total number of chromosomes.
* The **haploid (n) number** of chromosomes is half the diploid number.
* In humans, the diploid number of 46 is reduced to the haploid number of 23.
* **Gametes** (reproductive cells, often the sperm and egg) usually have the haploid number of chromosomes.
* Gamete formation and then fusion of gametes to form a cell called a **zygote** are integral parts of **sexual reproduction.** A **zygote** always has the full or diploid (**2n**) number of chromosomes. In plants and animals, the zygote undergoes development to become an adult organism.

(II) Homologous Pairs of Chromosomes:

In diploid body cells, the chromosomes occur in pairs. A pictorial display of human chromosomes shows the chromosomes arranged according to pairs. The members of each pair are called homologous chromosomes.

* **Homologous chromosomes** look alike; they have the same length and centromere position.



These chromosomes are duplicated, and each one is composed of two chromatids. The sister chromatids contain the exact same genes; the nonsister chromatids contain genes for the same traits (e.g., type of hair,

color of eyes), but they may differ in that one could "call for" dark hair and eyes and the other for light hair and eyes

Why does the zygote have two chromosomes of each kind?

***** One member of a homologous pair was inherited from the male parent, and the other was inherited from the female parent by way of the gamete

(III) Overview of Meiosis:

* Meiosis requires **two** nuclear divisions and produces **four** haploid daughter cells, each having one of each kind of chromosome.

* Replication occurs only once and the daughter cells have half the total number of chromosomes as were in the diploid parent nucleus.

* The daughter cells receive one of each kind of parental chromosome, but in different combinations.

* Therefore, the daughter cells are not genetically identical to the parent cell or to each other.

1- The phases of meiosis I:

(I) Prophase I

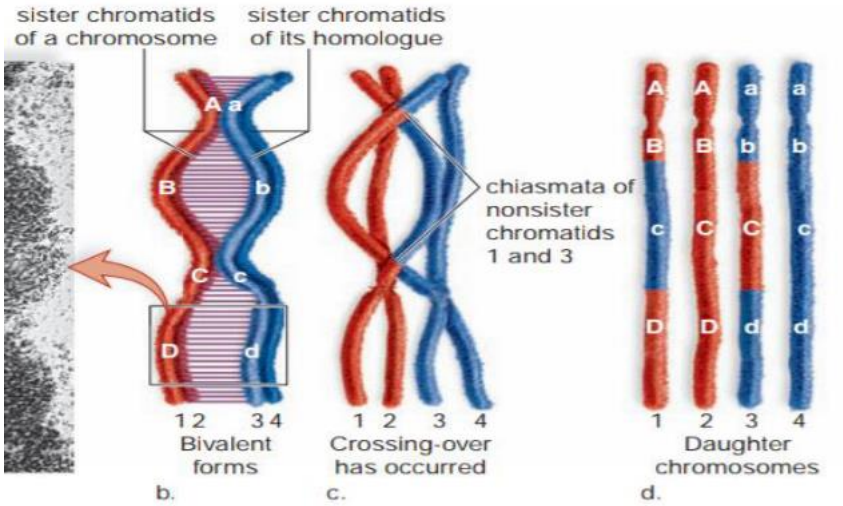
* A spindle forms as the centrosomes migrate away from one another.

* The nuclear envelope fragments, and the nucleolus **disappears**.

* The **homologous chromosomes**, each having two sister chromatids, undergo **synapsis** to form bivalents (tetrad; contains four chromatids).

* **Crossing-over (**is an exchange of genetic material between non-sister chromatids of a bivalent during meiosis I) may occur at this time.

* **Chiasmata** (sing., chiasma), regions where the non-sister chromatids are attached due to crossing-over.



(II) Metaphase I:

* During metaphase I, the bivalents held together by chiasmata have moved toward the metaphase plate (equator of the spindle).
* Metaphase I is characterized by a fully formed spindle and alignment of the bivalents at the metaphase plate.
* As in mitosis, kinetochores are seen, but the two kinetochores of a duplicated chromosome are attached to the same kinetochore spindle fiber.

(III) Anaphase I

* During anaphase I, the homologues of each bivalent separate and move to opposite poles, but sister chromatids do not separate. Therefore, each chromosome still has two chromatids.

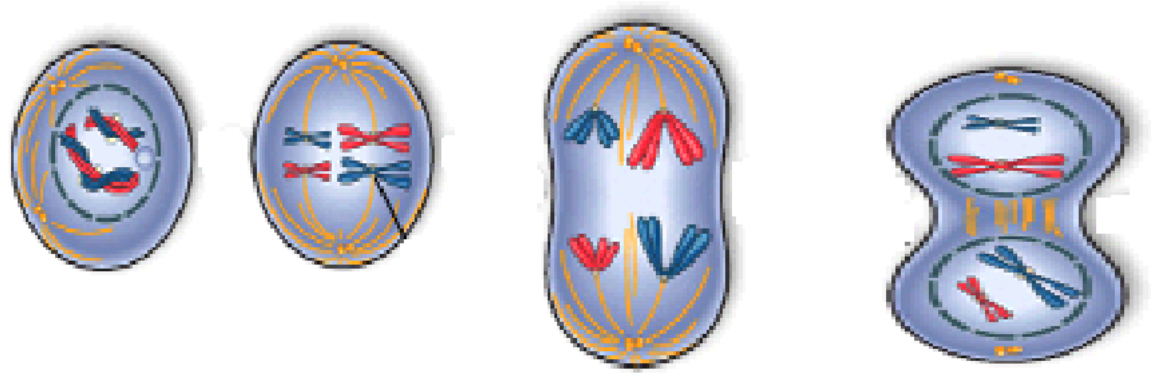
(IV) Telophase I

* Completion of telophase I is not necessary during meiosis.That is, the spindle disappears, but new nuclear envelopes need not form before the daughter cells proceed to meiosis II.
* Also, this phase may or may not be accompanied by cytokinesis, which is separation of the cytoplasm.

Interkinesis

* Following telophase, the cells enter interkinesis. The process of **interkinesis** is similar to interphase between mitotic divisions except that DNA replication does not occur because the chromosomes are already duplicated.

Meiosis I



Prophase I **Metaphase I** **Anaphase I** **Telophase I**

2- Meiosis II and Gamete Formation

(I) At **prophase II**, the two daughter cells contain the haploid number of chromosomes, or one chromosome from each homologous pair.

* Note that these chromosomes still consist of duplicated sister chromatids at this point.

(II) During **metaphase II**, the chromosomes align at the metaphase plate, but do not align in homologous pairs as in meiosis I because only one chromosome of each homologous pair is present.

* Thus, the alignment of the chromosomes at the metaphase plate is similar to what is observed during mitosis.

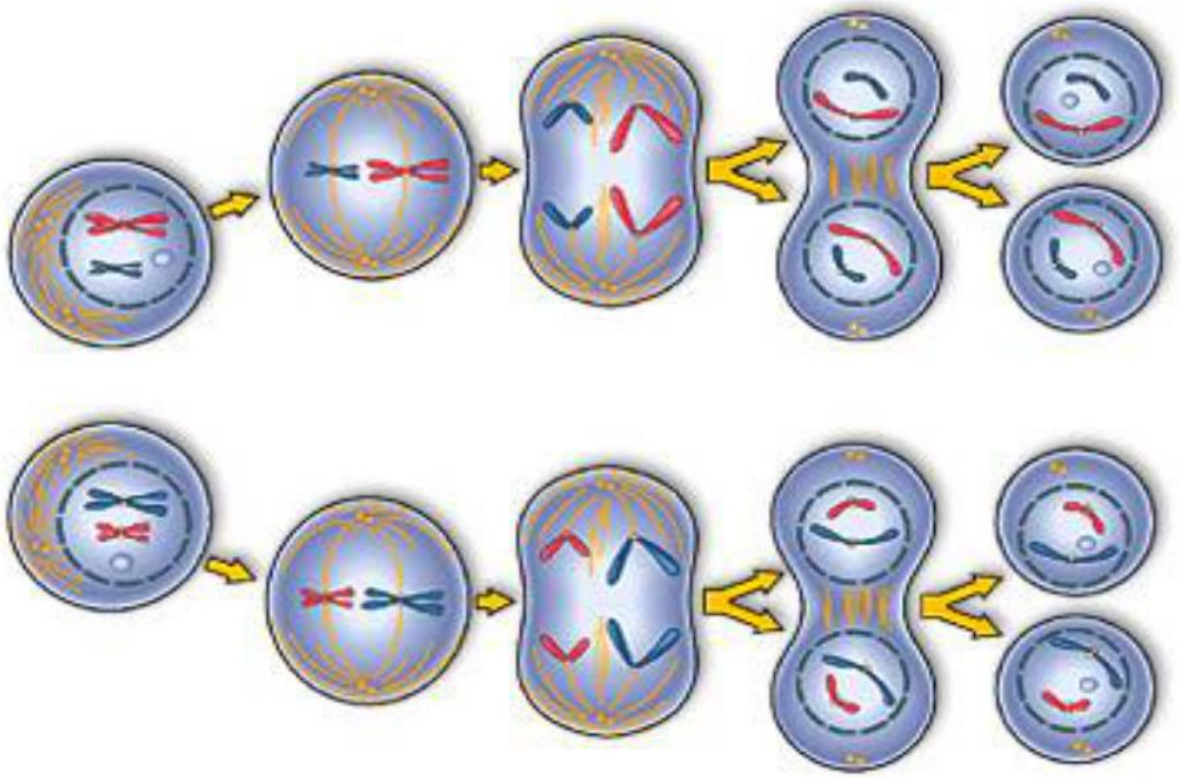
(III) During **anaphase II**, the sister chromatids separate, becoming daughter chromosomes that are not duplicated.

* These daughter chromosomes move toward the poles.

(IV) At the end of **telophase II** and **cytokinesis II**, there are four haploid cells.

* Due to crossing-over of chromatids during **meiosis I,** each gamete will most likely contain chromosomes with **varied** genes. As mentioned, following **meiosis II**, the haploid cells become gametes in animals. In plants, they become **spores.**

Meiosis II



ProphaseII MetaphaseII AnaphaseII TelophaseII CytokinesisII